INDIVIDUAL PROJECT REPORT

PROJECT 03

AI in Healthcare: Apollo Hospitals, India

Using AI to enable early disease detection, personalised treatment support, fewer diagnostic errors and better use of hospital resources

Student Details

Enrollment Number	Name
20261014	Devansh Kavishwar

The patient case used in this report (Sunita) is hypothetical and not a real case.

Abstract

India has roughly a quarter of the world's tuberculosis (TB) cases, and plenty of people with symptoms live a long way from anyone who can read an X-ray or order the right test. This report looks at how artificial intelligence (AI) could help the health system catch disease earlier, support safer and more personalised decisions, and make better use of scarce resources. I use TB as the worked example and rely only on public evidence. Some of the pieces are already in place: AI reading of chest X-rays is used in India's TB programme, and the Ayushman Bharat Digital Mission (ABDM) has linked about 104 crore health records. What I could not find is public evidence that these pieces work together as one loop covering triage, referral, follow-up and planning, or any public baseline for how long people wait for a test or whether they finish follow-up. The report proposes a six-stage loop with humans in charge, matches AI techniques to tasks, lays out a five-phase plan that starts by measuring, and sets out safeguards for six risks. Sunita, the patient used throughout, is hypothetical.

1. Introduction

Take Sunita, a hypothetical 42-year-old farm worker. She has had a cough for three weeks, she is losing weight, and she gets a fever in the evenings. The health centre nearest to her has no radiologist. It could be TB, pneumonia or chronic obstructive pulmonary disease (COPD), and nobody close by can say which. Any X-ray she has will sit and wait for a specialist. Her records are on paper or spread across separate systems, and each referral means another journey for her.

The brief asks how AI can help India's healthcare system detect disease earlier, support safer and more personalised clinical decisions, and use limited resources more effectively. The focus is fewer diagnostic errors, better outcomes and smarter use of resources across the whole health system, not within one hospital. I use Apollo Hospitals as a reference point because it already runs clinical AI at scale, but the problem belongs to the system and not to one provider. The question I kept coming back to was simple: how do we move from reacting to illness towards finding it early and following it through?

1.1 Approach and scope

This is a desk study. I used peer-reviewed papers, official statistics, regulatory documents and company announcements, and I did not have access to any hospital or programme data. I also did not test a model myself. Wherever I describe something, I say whether it exists today (and can be checked in a public source) or whether it is our proposal. Where I could not find a figure, I say that and leave it out rather than guess. The report moves from the evidence (Section 2) to what exists (Section 3) and the gap (Section 4), then to the proposed system (Section 5), the techniques (Section 6), an example (Section 7), benefits (Section 8), evaluation (Section 9), the roadmap (Section 10), risks (Section 11), ethics and equity (Section 12), and finally limitations and a conclusion. Charts and tables are in the Appendix.

2. The Problem: What the Evidence Shows

2.1 A heavy burden

About 25% of the world's TB cases are in India (2024), and so is 32% of drug-resistant TB (WHO, 2025; Figure A1). Things are improving, since incidence has fallen 21% since 2015. Still, an estimated one lakh cases in 2024 never reached the national programme. Those are people like Sunita: sick, but never properly inside the system.

2.2 Too few specialists where they are needed

Rural Community Health Centres (CHCs) have a 79.5% shortfall of specialists, and just 541 of 5,480 CHCs, roughly 9.9%, have all four basic specialists (Ministry of Health and Family Welfare, 2022; Figure A2). This data stops at March 2022, so it is older than I would like. But the shortage is structural and I doubt it has gone away. With no radiologist, an X-ray either waits for a reader somewhere else or the patient has to travel to one.

2.3 The tools exist but are not joined up

The ABDM has linked about 104 crore health records to more than 93 crore ABHA accounts as of July 2026, which is a real national backbone for records. AI reading of chest X-rays is already part of India's TB programme, and the World Health Organization has recommended computer-aided detection for people aged 15 and above (WHO, 2021). So the ingredients are there. What I could not find was public evidence of them working together as one loop for triage, referral, follow-up and planning. That is my own reading of the sources. Not finding something is not proof that nobody is doing it.

2.4 Why the layers need to be connected

Sunita's case involves three layers: the patient (is this probably TB?), the doctor (what should be tested next?) and the health system (are kits, lab time and follow-up actually available?). When one layer fails, the others feel it. An X-ray read late leads to a late test, then possibly a patient who drops out, and a longer period during which others can be infected. Improving one layer on its own leaves a lot of the benefit unused. This is reasoning, not a measured result, but it is the main reason I argue for a loop rather than separate tools.

3. What Exists Today (Verified)

Three things exist already. AI X-ray reading is part of the TB programme. ABDM gives a records backbone with consent controls. And private-sector products show the supporting technology is mature. I mention three examples only to show the building blocks are real: Apollo's AICVD cardiac risk model, Siemens Healthineers' AI-Rad Companion and Practo's ABDM-compliant records software. They are not part of the proposal and I am not endorsing them.

AICVD is worth a closer look because it was built on Indian data. Bermejo et al. (2024) report that it was developed on 31,599 participants from six Apollo hospitals, and in validation it reached 80.15% accuracy compared with 59.71% for the Framingham score and 51.57% for QRisk3 (Figure A7). It is a heart model, not a TB tool, and the results are from retrospective validation. I cite it for one reason. It shows that models can be built and checked on Indian patients, and that scores designed for other populations do not simply carry over. How such a model behaves in a busy live clinic is a different question, which the evaluation plan in Section 9 is meant to answer.

4. The Gap

The gap described here is my own analysis, not a statement from any programme or company, and it rests on what I could and could not find publicly. The first gap is integration. I found no public evidence of AI X-ray reading, linked records, referral, follow-up and resource planning working together as one connected loop. The second is baselines. I found no reliable public figures for the days between a first visit and a confirmatory test, the share of people who complete follow-up, or X-ray and lab waiting times. Without a baseline you cannot show that anything improved. So the first recommendation has to be to measure where things stand before deploying anything new.

5. Proposed System: One Connected Loop

I propose one loop rather than three separate tools (Figure A3). It starts with patient and provider data: the X-ray, symptoms, labs and records, linked through ABHA with consent. Next come secure quality checks, which fix missing data, match records that belong to the same person and control who can see what. The third stage holds the models and knowledge: image AI, rules, a knowledge graph and forecasts. These produce alerts and suggestions, such as a risk flag, a next test or a resource forecast. Then comes the stage that matters most, humans decide. The doctor and health team accept, change or reject each suggestion. The last stage monitors accuracy, fairness, drift and real outcomes and feeds what it finds back into the models.

Three modules run on the loop (Table A1). Module A is early detection. The AI reads a chest X-ray and gives a TB-likelihood score, and a high score means the person needs a confirmatory test, not that they have TB. This already exists in the programme, so what we add is linked records and a doctor's review. Module B is coordinated care. Programme rules, a knowledge graph and a measure of uncertainty suggest the next test and the follow-up, and the doctor decides. Putting rules, graph and follow-up tracking in one place is new. Module C is resource planning, meaning forecasts of X-ray, lab and test-kit demand and scheduling of referrals and follow-up across facilities, which is also new. I would start with TB in a few districts. If the evidence supports it, the same loop could later be used for conditions such as diabetes or heart disease.

5.1 Design principles

Five ideas shaped the design. A human stays in charge, so every alert is advice that a clinician can accept, change or override. Each suggestion shows its main reasons and its uncertainty, so a doctor can judge it instead of just following it. No model is trusted because it did well somewhere else, so everything is validated locally. Only the data a task needs is used, access depends on role and every access is logged. And monitoring is part of the design from day one, not something added later.

5.2 Why one loop and not three tools

Three separate tools would each need their own data feed, alert rules and monitoring, and clinicians would get three unconnected streams of advice. In one loop the same cleaned data feeds every module, the same rules govern every alert, and a single monitoring stage can pick up bias or drift anywhere. A rise in high-risk X-ray scores from Module A can also feed the demand forecast in Module C, so test kits and referrals are planned earlier. That link between modules is exactly the gap from Section 4.

6. The AI Techniques Behind the System

I tried to match each task to the technique that suits it, rather than reaching for whatever is fashionable (Table A2). Between them the techniques cover learning from data, reasoning under uncertainty, representing knowledge and generating text, with a governance layer wrapped around the lot.

6.1 Deep learning and supervised machine learning (learn)

A deep neural network reads the X-ray and produces the TB-likelihood score for Module A. How accurate it is depends on the setting and on the threshold chosen, so it has to be tested locally. Supervised machine learning learns from labelled past cases to estimate risk and forecast demand in Modules A and C. Its weak spot is data. It needs representative examples, including rural patients, who are the people least likely to be in hospital datasets. For forecasting, studies from other settings suggest not assuming that complex means better. One

study of eight years of admissions data from two London hospitals found linear models often did better than machine learning methods (Vollmer et al., 2021), and an NHS Trust study found ARIMA forecasts beat the Trust's existing model over six weeks (Eyles et al., 2022). Those are admission studies and not TB studies, so I take them only as a reason to pick the forecasting method by testing on local data.

6.2 Probabilistic reasoning and expert systems (reason)

Probabilistic reasoning lets the system show a probability with its uncertainty instead of a flat yes or no. The scores need to be calibrated, so that a stated 18% really means about 18 in 100 patients, and doctors should be careful not to over-trust them. An expert system applies explicit programme rules, for example that a high score means a confirmatory test and a notification. Rules are easy to audit but rigid and cannot learn new patterns, so they are paired with the learning models.

6.3 Knowledge graph and ontology (represent)

A knowledge graph with an ontology links symptoms, tests, drugs and facilities using a shared vocabulary. It supports Module B and lets the system explain why it suggests a particular test. The downside is cost, since these graphs are expensive to build and to keep up to date.

6.4 Generative AI (generate)

Generative AI would draft referral summaries, using approved records only. The risk is that it invents or drops details. One large evaluation of clinical note generation found a 1.47% hallucination rate and a 3.45% omission rate (Asgari et al., 2025; Figure A5). Those numbers look small, but across a whole health system one wrong or missing detail can matter, so a doctor always checks a draft before it is used.

6.5 Responsible AI (govern)

Responsible AI sits around everything else: bias checks, privacy protection, audit logs and human sign-off. It costs time and needs oversight, and I treat that as part of the system and not an optional extra. All the AI in this report is narrow AI, meaning it is built for specific tasks and is not general intelligence.

7. Worked Example: Sunita's Journey (Hypothetical)

Table A3 follows Sunita through the system. At the village visit, a health worker records her symptoms and takes a digital X-ray, and the record is linked to her ABHA ID with her consent. This part exists today. Then the image AI gives a TB-likelihood score along with its uncertainty, and the rules say a high score should lead to a confirmatory test (our proposal). A district doctor looks at the X-ray, the score and her history, can override the suggestion, and in this case orders a molecular test (our proposal). Finally the referral is booked, reminders go out, and results are shared with her consent with her local clinic, so someone near home can follow up (our proposal).

At the same time, district planners see forecasts of X-ray, lab and test-kit demand and of follow-up gaps, which helps them send vans, staff and kits where they are needed. The AI never diagnoses or treats anyone on its own. If it works, Sunita's X-ray is read sooner, a doctor decides what comes next and somebody follows up. AI helps with that chain, but it cannot remove error or guarantee an outcome.

8. Expected Benefits by Stakeholder

These benefits are expected, not demonstrated. Patients could be picked up earlier, tested sooner and told clearly why a test is needed, and they would not have to repeat their history

as often because it is in one place. Health workers would get a simple way to record, screen and refer, and a clearer idea of what happens after a referral. Doctors would see the X-ray, score and history together and could put their limited time where it is needed most, though extra alerts are a real risk for them, which is why alert counts are part of the evaluation. District planners would have forecasts instead of guesswork for X-rays, labs and test kits. Programme leaders would get measured baselines, an audit trail and evidence for deciding whether to expand. Section 9 explains how each of these would be tested.

9. How We Would Know It Worked: Evaluation Plan

Since public baselines are missing, the plan starts with measuring. There are five phases (Figure A6). Phase 1 defines the use case and partners and measures today's numbers. Phase 2 builds: choose the model and test it on past data. Phase 3 is a pilot in a few facilities with doctors in charge. Phase 4 evaluates safety, fairness, workflow and cost, and Phase 5 expands only if the evidence supports it. Before any scores are shown to doctors, I would also let the model run quietly beside usual care and compare its predictions with what really happened.

The plan asks four plain questions (Table A5). First, do we catch real TB? Of the people who truly have TB, how many are flagged, and how many false alarms come with that? For this I use the WHO triage benchmark of at least 90% sensitivity and at least 70% specificity (WHO, 2014; Figure A4). Sensitivity is the share of true cases that get flagged, and specificity is the share of people without TB who are correctly left alone. Second, are people tested sooner, measured as days from first visit to confirmatory test? Third, do people finish follow-up, measured as the share who complete referral and follow-up visits? Fourth, are resources used better, judged by X-ray and lab waiting times and by forecast error compared with today's method.

There is no reliable public baseline for the second, third and fourth questions, so Phase 1 has to measure them first. Each result is then compared with the same site's own figures from before, which is fairer than comparing one district with another. The success criteria, including how many alerts is acceptable, are agreed with clinicians before the pilot starts, so nobody can read the results to suit the outcome.

10. Implementation Roadmap

The roadmap is my proposal, and I give no calendar durations because realistic timelines depend on the baselines measured first. Phase 1 should also set up governance: a small group of doctors, data specialists, IT security, a privacy officer and programme managers, with accountability, consent procedures and audit logging agreed before any model goes live. After that each phase depends on the one before. Pilots start only after the model has been tested on past data, expansion follows an honest evaluation, and the decision to expand is judged against criteria set in advance, such as no fall in safety, an acceptable alert load and accuracy within the agreed tolerance.

Costs and limits should be stated plainly. The main cost areas are devices, connectivity, staff training, maintenance and regulatory approval. I quote no rupee figures because I could not find reliable public ones. Many facilities are not fully digital, so I would start where basic digital X-ray already exists and widen from there as the digital foundations improve.

11. Risks and Safeguards

I tried to plan for failure first, so each of the six main risks has a safeguard and someone responsible (Table A4). Biased or thin data is handled by validating by region, sex and age and by deliberately including rural and underserved patients. False alarms and missed cases are tracked, and a confirmatory test and a doctor make the final call, so the AI score never

decides alone. To limit over-reliance, the system shows uncertainty and reasons, the doctor can override, and every override is logged. Privacy and consent go through ABDM consent and access controls. For the rules on medical AI, the project would follow CDSCO guidance on medical device software and the ICMR ethical guidelines. Model drift is handled by continuous monitoring, with retraining and retesting when accuracy slips.

12. Ethical, Legal and Equity Considerations

Fairness here means that no patient is worse off because of age, sex, income, region or any other attribute, which is why validation is split by group. Legally, the project has to follow the Digital Personal Data Protection Act, 2023 and its Rules, which come fully into force by May 2027, along with the CDSCO guidance for software that acts as a medical device. The ICMR (2023) guidelines on AI in biomedical research and healthcare set out ethical expectations such as consent, transparency and accountability.

Equity needs particular attention. If we start only where digital X-ray exists, the first benefits will probably reach better-equipped districts, so the roadmap should deliberately widen to underserved areas. TB can also carry stigma, so reminders and shared results must respect privacy and the patient's say over who is told what.

13. Limitations and Recommendations

The main limitations are easy to list. This is a desk study and I tested no model. Some figures are older than I would like, such as the specialist data to March 2022. Several measures have no public baseline. And the proposal is unproven, so there is no guarantee it would work. My recommendation is to go carefully: measure the current position first, begin with TB in a few districts where digital X-ray already exists, validate locally and independently, keep humans in charge of every decision, and publish the results, including any failures.

14. Conclusion

India already has several of the pieces needed to catch disease earlier: AI X-ray reading, a national records system and a growing set of tested tools. My argument is that the next gain comes from joining them into one loop, with humans in charge and honest measurement at every stage. For Sunita that could mean an X-ray read sooner, a doctor who decides and someone who follows up. AI supports that chain but cannot remove error or guarantee outcomes, so the idea should be piloted carefully and expanded only on proof.

References

- Apollo Hospitals (n.d.). AICVD cardiac risk model and clinical AI programme [Company information]. Apollo Hospitals Enterprise Ltd. https://www.apollohospitals.com (accessed October 2026).

Asgari, E., et al. (2025). A framework to assess clinical safety and hallucination rates of LLMs for medical text summarisation. npj Digital Medicine, 8.

Bermejo, I., Gupta, P., Jalepalli, S. K., Dekker, A. L. A. J., & Kar, S. (2024). Development and validation of multicentre study on novel artificial intelligence-based cardiovascular risk score (AICVD). Family Medicine and Community Health, 12(Suppl 1), e002340. https://doi.org/10.1136/fmch-2023-002340

Central Drugs Standard Control Organisation (n.d.). Guidance on medical device software. CDSCO, New Delhi.

Eyles, E., Redaniel, M. T., Jones, T., Prat, M., & Keen, T. (2022). Can we accurately forecast non-elective bed occupancy and admissions in the NHS? A time-series MSARIMA analysis of longitudinal data from an NHS Trust. BMJ Open, 12(4), e056523.

Government of India (2023). The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023). Ministry of Law and Justice, New Delhi.

Government of India (2025). Digital Personal Data Protection Rules, 2025. Ministry of Electronics and Information Technology, New Delhi.

Indian Council of Medical Research (2023). Ethical guidelines for application of artificial intelligence in biomedical research and healthcare. ICMR, New Delhi.

Ministry of Health and Family Welfare (2022). Rural Health Statistics 2021-22. Government of India, New Delhi.

National Health Authority (2026). Ayushman Bharat Digital Mission: ABHA accounts and linked health records (figures as of July 2026). https://abdm.gov.in (accessed October 2026).

Practo (n.d.). ABDM-compliant records software [Product information]. Siemens Healthineers (n.d.). AI-Rad Companion [Product information]. Cited only as examples of existing building blocks.

Vollmer, M. A. C., Glampson, B., Mellan, T. A., Mishra, S., Mercuri, L., Costello, C., Klaber, R., Cooke, G., Flaxman, S., & Bhatt, S. (2021). A unified machine learning approach to time series forecasting applied to demand at emergency departments. BMC Emergency Medicine, 21. https://doi.org/10.1186/s12873-020-00395-y

World Health Organization (2014). High-priority target product profiles for new tuberculosis diagnostics: report of a consensus meeting. WHO, Geneva.

World Health Organization (2021). WHO consolidated guidelines on tuberculosis. Module 2: screening. Systematic screening for tuberculosis disease. WHO, Geneva.

World Health Organization (2025). Global tuberculosis report 2025. WHO, Geneva.

Appendix

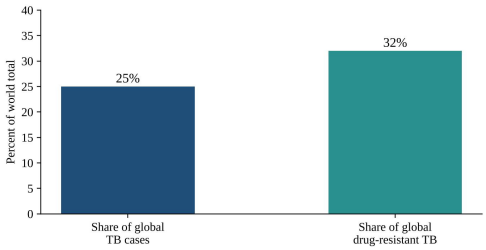


Figure A1. India's share of the world's TB cases and drug-resistant TB cases (2024). Source: WHO (2025).

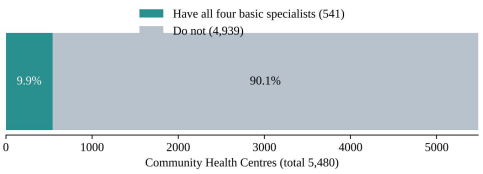


Figure A2. Community Health Centres with all four basic specialists (data to March 2022). Source: Ministry of Health and Family Welfare (2022).

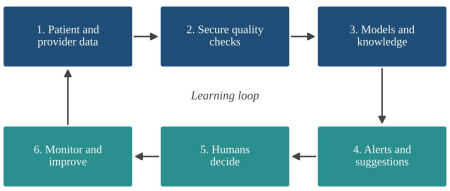


Figure A3. The proposed connected loop, from patient data to monitoring and back. The human decision is stage 5.

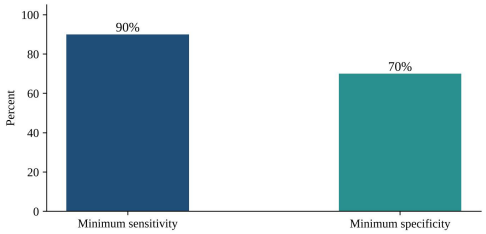


Figure A4. WHO triage benchmark for TB screening tools. Source: WHO (2014).

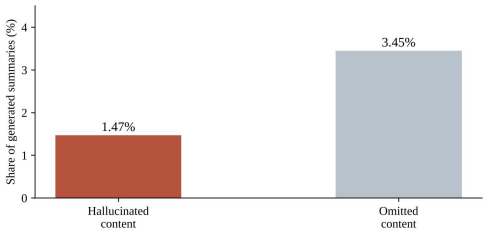


Figure A5. Error rates reported in one study of AI-generated clinical text. Source: Asgari et al. (2025).



Figure A6. The five-phase plan, from defining the use case to expansion.

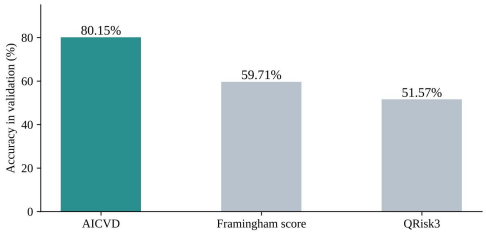


Figure A7. Accuracy of AICVD compared with the Framingham score and QRisk3 in validation. Source: Bermejo et al. (2024).

Table A1. Modules: what exists today versus our proposal

Module	Exists today (verified, public)	Our proposal
A. Early detection	AI chest X-ray reading is part of India's TB programme.	Link X-ray results to records and add doctor review; a high score prompts a test, not a diagnosis.
B. Coordinated care	ABDM provides records and consent controls.	Unified rules plus knowledge graph suggest the next test and follow-up; the doctor decides.
C. Resource planning	Not publicly evidenced across facilities.	Forecast X-ray, lab and test-kit demand; schedule referrals and follow-up; method chosen by local testing.
Integrated loop	Not publicly evidenced.	Six-stage loop with monitoring of accuracy, fairness, drift and outcomes.

Table A2. Technique mapping

Technique	Role	Use in this system	Main limitation
Deep learning	Learn	Reads the X-ray and outputs a TB-likelihood score (Module A).	Accuracy varies by setting and threshold; test locally.
Supervised ML	Learn	Estimates risk and forecasts demand (Modules A and C).	Needs representative data, including rural patients.
Probabilistic reasoning	Reason	Shows a probability with uncertainty, not yes or no.	Needs calibrated scores; can be over-trusted.
Expert system (rules)	Reason	High score means confirmatory test and notification.	Rigid; cannot learn new patterns.
Knowledge graph and ontology	Represent	Links symptoms, tests, drugs and facilities (Module B).	Costly to build and keep current.
Generative AI	Generate	Drafts referral summaries from approved records only.	1.47% hallucinated and 3.45% omitted in one study; doctor verifies.
Responsible AI	Govern	Bias checks, privacy, audit logs, human sign-off.	Requires time, cost and oversight.

Table A3. Sunita's journey (hypothetical)

Step	Stage	What happens	Status
1	Village visit	Health worker records symptoms and takes a digital X-ray; record linked to her ABHA ID with consent.	Exists today
2	AI flag	Image AI gives a TB-likelihood score with uncertainty; rules: high score, send for confirmatory test.	Our proposal
3	Doctor review	District doctor sees X-ray, score and history, can override, and orders a molecular test.	Our proposal
4	Follow-up	Referral booked, reminders sent, results shared with consent to her local clinic.	Our proposal
Plan	District planners	Forecasts of X-ray, lab and test-kit demand and follow-up gaps guide vans, staff and kits.	Our proposal

Table A4. Risks and safeguards

Risk	Safeguard
Biased or thin data	Validate by region, sex and age; include rural and underserved patients.
False alarms, missed cases	Track false alarms and misses; a confirmatory test and a doctor decide.
Over-reliance on AI	Show uncertainty and reasons; doctor can override; overrides are logged.
Privacy and consent	Consent and access controls through ABDM; follow the DPDP Act and Rules (fully in force by May 2027).
Rules for medical AI	Follow CDSCO medical-device-software guidance and ICMR 2023 AI ethics guidelines.
Model drift	Keep monitoring accuracy; retrain and retest when it slips.

Table A5. Evaluation plan: four questions

Question	Metric	Comparison
Do we catch real TB?	Sensitivity and false-alarm rate.	WHO triage benchmark: at least 90% sensitivity, 70% specificity.
Are people tested sooner?	Days from first visit to confirmatory test.	Same site's own baseline, measured in Phase 1.
Do people finish follow-up?	Share completing referral and follow-up visits.	Same site's own baseline, measured in Phase 1.
Are resources used better?	X-ray and lab waiting time; forecast error.	Today's method at the same site.